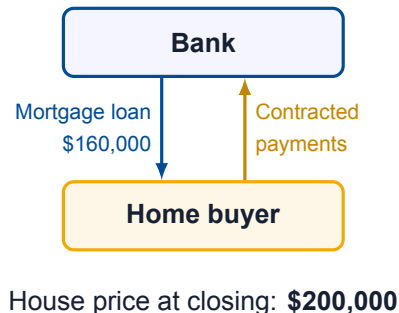


# A House, a Mortgage, and Inflation

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- ▶ The bank charges a fixed nominal interest rate of **8%**. Both sides expect **5% inflation**.
- ▶ The house price is fixed when the purchase closes. The loan contract determines what changes afterward.
- ▶ A **fixed nominal mortgage** does not automatically adjust its scheduled dollar payments for inflation.
- ▶ An **inflation-indexed mortgage** adjusts the balance or payments using an agreed price index.

**What happens when actual inflation differs from 5%?**

# Unexpected Inflation

**The mortgage example:** fixed nominal rate 8%; expected annual inflation 5%.

| Inflation    | Approximate real rate | Compared with expectations |
|--------------|-----------------------|----------------------------|
| Expected: 5% | $8 - 5 = 3\%$         | Expected outcome           |
| Actual: 6%   | $8 - 6 = 2\%$         | Borrower benefits          |
| Actual: 2%   | $8 - 2 = 6\%$         | Lender benefits            |

- ▶ Higher inflation reduces the purchasing power of the repayment relative to what the lender expected.
- ▶ Lower inflation increases that purchasing power. The fixed payment is a larger real burden for the borrower.